Flood Mapping of the October 2025 Pantepec River Flood in Álamo, Veracruz, Using Multispectral Imagery and Supervised Classification

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***Abstract*—In October 2025, extreme rainfall caused the Pantepec river to overflow, flooding the town of Álamo in northern Veracruz, Mexico. Rapid and reliable delineation of flooded areas is essential to support emergency response, damage assessment, and recovery planning. One of the current methodologies for this task is the United Nations UN-SPIDER recommended practice for flood mapping using Google Earth Engine, a change-detection approach based on Sentinel-1 Synthetic Aperture Radar (SAR) backscatter thresholding, combined with the JRC Global Surface Water dataset to mask permanent water bodies. This work proposes a complementary methodology based on Sentinel-2 multispectral imagery: pre- and post-event optical bands, together with derived spectral indices (MNDWI, NDWI, NDVI, NDBI, NDTI, AWEIsh), are used as features to train supervised machine learning classifiers. Two models, a Support Vector Machine (SVM) and a Random Forest (RF), are trained using the SAR-threshold flood mask as ground truth. Sequential Feature Selection was used in order to obtain the most significant bands together with a stratified random sample of pixels. Models are compared through their confusion matrices and cross-validated accuracy. Applied to the full study area, the optical classifiers detect a larger flooded extent than the threshold-based method. The results suggest that optical-based supervised classification can act as an alternative flood mapping tool when SAR acquisitions are unavailable, and as an independent cross-check of the traditional SAR threshold methodology.**

I. Introduction

Between the 6th and the 10th of October 2025, the residuals of tropical systems on the Gulf of Mexico produced extreme precipitation over the states of Veracruz, Puebla, Hidalgo, Querétaro and San Luis Potosí, causing one of the most severe flooding events recorded in the region [1]. In the north of Veracruz, the Pantepec river (which becomes the Tuxpan river downstream) overflowed its banks, inundating large portions of the municipality of Álamo Temapache. The flood affected thousands of homes, agricultural land (mainly citrus plantations, the economic backbone of the area) and critical infrastructure causing power outages that lasted for days.

One of the current operational reference methodology is the UN-SPIDER *Recommended Practice: Flood Mapping and Damage Assessment Using Sentinel-1 SAR Data in Google Earth Engine* [2], with a demonstrated use-case on the March 2019 flooding caused by cyclone Idai in Mozambique. It is a change-detection approach: the ratio between the post-event and pre-event SAR backscatter mosaics is thresholded to obtain a binary flood mask, and the JRC Global Surface Water dataset [3] is used to remove permanent water bodies (areas covered by water more than 10 months per year). SAR is preferred for flood response because it is agnostic to cloud cover and daylight; however, the method depends on the availability of suitable pre/post acquisitions on the same orbit, on a manually tuned threshold, and provides no measure of classification confidence.

This work proposes to complement the SAR threshold methodology with Sentinel-2 optical imagery. Pre- and post-event surface reflectance bands and a set of derived spectral indices sensitive to water, vegetation, built-up surfaces and turbidity (MNDWI, NDWI, NDVI, NDBI, NDTI, AWEIsh [4]) are combined into a multi-temporal feature collection. As a first assessment of this methodology, a Sequential Feature Selection tool is used to extract the most important bands and spectral indices, together with two supervised classifiers, a Support Vector Machine and a Random Forest, trained to detect flooded pixels using the UN-SPIDER SAR result as ground truth for the training samples. The advantage of this approach is twofold: (i) it provides a flood detection tool for cases in which SAR data are not available or not usable for a given event, and (ii) it acts as a verification step of the threshold-based product, since the two data sources rely on completely different physical principles (microwave backscatter vs. solar reflectance).

Figure 1 shows the Area of Interest (AOI), a tight bounding region around the town of Álamo, Veracruz, Mexico (∼21°N, 97.7°W).

II. Data

Two satellite data sources, both accessed through the Google Earth Engine (GEE) [10] catalog, were used.

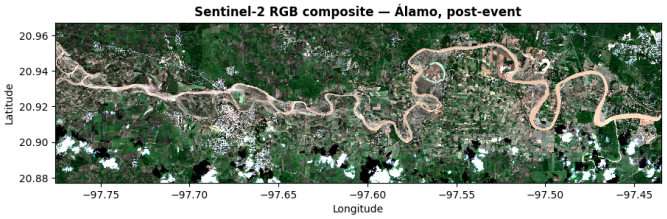


Fig. 1. RGB composite of the Area of Interest around Álamo, Veracruz.

1) Sentinel-1 SAR: The Sentinel-1 Ground Range Detected (GRD) collection (`COPERNICUS/S1_GRD`) was used to reproduce the UN-SPIDER reference product. C-band SAR is an active sensor: it measures the backscatter of a transmitted microwave pulse, so it observes the surface through clouds and at night, which is decisive during a flood, when persistent cloud cover typically prevents optical acquisitions. Smooth open water acts as a specular reflector and returns very low backscatter, so newly flooded land appears as a strong drop in the backscatter ratio. The following acquisition parameters were fixed:

- Instrument mode: Interferometric Wide swath (IW), 10 m resolution;
- Polarization: VH (cross-polarization, preferred for flood mapping over VV);
- Orbit: descending pass, relative orbit number 143, so that pre- and post-event scenes share the same viewing geometry.

2) Sentinel-2 multispectral: The Harmonized Sentinel-2 Level-2A surface reflectance collection (`COPERNICUS/S2_SR_HARMONIZED`) was used for the proposed optical classification. Level-2A provides atmospherically corrected bottom-of-atmosphere reflectance together with quality/cloud probability layers. All spectral bands from B1 to B12 (visible, red edge, NIR and SWIR; native resolutions of 10–60 m/px) were retained as features, and six spectral indices were added (Section III-A). Optical data are passive measurements of reflected sunlight: they offer a rich spectral characterization of the surface (water absorption in the NIR/SWIR, vegetation red edge, turbidity in the visible bands) that a single-band SAR ratio cannot provide, but they are blocked by clouds. Using both sources therefore combines the reliability of SAR with the spectral richness of optical imagery.

For both sensors, two time windows were defined around the event: a *before* period from 2025-09-01 to 2025-10-05 and an *after* period from 2025-10-10 to 2025-10-15.

III. Materials and Methods

The workflow was implemented partly in Google Earth Engine (data selection, pre-processing, band algebra, sampling and export) and partly in a Python notebook (sequential feature selection, model training, hyperparameter tuning, raster inference and comparison), since Python's `scikit-learn` offers finer control over the classifier hyperparameters and evaluation than the GEE classification toolbox. The main libraries used were `Google Earth Engine` [10] `geemap` (GEE–Python bridge), `rasterio` (raster I/O), `numpy`, `pandas`, `matplotlib`/`seaborn` and `scikit-learn` [7].

A. Pre-processing

1) SAR: For each of the two date windows, the Sentinel-1 collection was filtered by date, AOI, IW mode, VH polarization, descending pass and relative orbit 143, mosaicked and clipped to the AOI. A circular focal-mean kernel of 50 m radius was applied to each mosaic to reduce the "salt and pepper" speckle noise inherent to SAR imagery. The flood proxy was then obtained as the ratio

$$D = \frac{\sigma^0_{\text{after}}}{\sigma^0_{\text{before}}}, \tag{1}$$

smoothed again with the same kernel and thresholded at $D > 1.25$ to produce the binary flood mask used as reference/ground truth.

2) Optical: The Sentinel-2 collections for the two windows were filtered by AOI and by cloud coverage (`CLOUDY_PIXEL_PERCENTAGE` $< 20\%$), and a per-pixel cloud mask based on the cloud probability layer (`MSK_CLDPRB` < 5) was applied before rescaling reflectance by the factor 10^{-4}. Six spectral indices were then added to each image, each justified by the surface property it enhances:

- **MNDWI** $= (G - SWIR1)/(G + SWIR1)$: modified water index, better separation of water from built-up areas, relevant for an urban flood;
- **NDWI** $= (G - NIR)/(G + NIR)$: classical open-water index;
- **NDVI** $= (NIR - R)/(NIR + R)$: vegetation vigor; flooding suppresses the vegetation signal of inundated fields and provides high contrast on water bodies.;
- **NDBI** $= (SWIR1 - NIR)/(SWIR1 + NIR)$: built-up index, helps discriminate urban surfaces from water and soil;
- **NDTI** $= (R - G)/(R + G)$: turbidity index; flood water is heavily sediment-laden, so turbidity is a strong signature of fresh flooding;
- **AWEIsh** $= B + 2.5\,G - 1.5\,(NIR + SWIR1) - 0.25\,SWIR2$: automated water extraction index with shadow suppression, robust in low-albedo urban shadows.

The indices were selected from the Awesome Spectral Indices catalog [4]. Each filtered collection was reduced to a single mosaic per window and clipped to the AOI.

Rather than feeding the classifier both dates separately, the per-band *difference* image (before − after) was computed for all 12 reflectance bands and 6 indices, yielding an 18-dimensional feature vector per pixel. This halves the dimensionality of the dataset and encodes the change information directly, which is what actually characterizes a flood. The SAR VH band was deliberately excluded from the feature stack: including it would let the model overfit to the very variable

used to generate the labels, and the goal of the experiment is to provide a tool for when SAR is *not* available.

B. Sampling

A stratified random sample of 2000 points per class (flooded / not flooded, 4000 points in total) was drawn at 10 m scale, using the SAR binary mask as the class band and a fixed seed for repeatability. Stratification is necessary because the dataset is strongly imbalanced: only ∼13% of the AOI is flooded according to the SAR mask, so a purely random sample would under-represent the flood class and bias the estimators. The sample size of 2000 points per class was considered adequate for a binary problem over an AOI of ~ 3.8 million pixels, keeping the GEE export within platform limits. Point geometries were *not* exported: discarding coordinates prevents any spatial information from leaking into the model and reduces the tendency to overfit to specific regions of the image, favoring a more general classifier. The samples were randomly split into 80% training and 20% validation, following common practice in the remote sensing classification literature, and exported to a pandas DataFrame.

C. Support Vector Machine

The SVM is well suited to binary problems with a moderate number of informative features: it finds the hyperplane that maximizes the margin between the two classes, using an RBF kernel to handle non-linear boundaries. The model was implemented as a `scikit-learn` pipeline of a `StandardScaler` (feature scaling is critical for SVM convergence) followed by an `SVC` with RBF kernel.

1) Forward Feature Selection: A very rustic, but informative method of extracting the features that most inform the classification model. Provided an SVM classification model, prioritizing for *accuracy* and stopping the feature aggregation after the increments are below 0.1% It reduced the number of features to 8. Order does not reflect significance:

B1, B9, B11, B12, MNDWI, NDWI, NDVI, AWEIsh

2) Hyperparameter selection: A grid search over $C \in \{0.1, 1, 10, 100, 1000\}$ and $\gamma \in \{\text{scale}, 0.0001, 0.001, 0.01\}$ was performed with a stratified 5-fold cross-validation on the training set. Stratified K-fold is used so that every fold preserves the 50/50 class ratio of the stratified sample; with a plain K-fold, random folds could end up class-imbalanced and yield misleading accuracy estimates. The best configuration was $C = 1$, $\gamma = scale$, with a cross-validated accuracy of 0.869 ± 0.013. Table I reports the confusion matrix on the held-out validation set (509 points); the overall validation accuracy was 0.826 with precision 0.826 and recall 0.944 for the flood class.

D. Random Forest

Random Forest is an ensemble of decision trees, each trained on a bootstrap sample of the data and a random subset of the features. Averaging many decorrelated trees reduces variance and makes RF robust to the noise (residual clouds, mixed pixels) which is the scenario for this case; it also requires no feature scaling/normalization.

TABLE I
CONFUSION MATRIX, SVM CLASSIFIER (VALIDATION SET)

	Predicted no flood	Predicted flood
Actual no flood	342	79
Actual flood	22	374

1) Forward Feature Selection: Using a forest classifier of 100 estimators and making sure the results of bootstrapping are reproducible with a fixed random state, it obtained the following list of 7 features. Order does not reflect significance:

B1, B5, B8A, B9, B12, MNDWI, NDWI

2) Hyperparameter selection: Values selected for the actual forest classification via a preliminary grid search over $n_{\text{estimators}} \in \{100, 200, 300\}$, depth $\in \{\text{None}, 10, 20\}$]. Model stability was assessed with the same set of stratified k-fold cross-validation used for SVM on the training set for unbiased comparison, obtaining an accuracy of 0.869 ± 0.018. Table II reports the confusion matrix on the validation set; the overall validation accuracy was 0.834, with precision 0.834 and recall 0.927 for the flood class.

TABLE II
CONFUSION MATRIX, RANDOM FOREST CLASSIFIER (VALIDATION SET)

	Predicted no flood	Predicted flood
Actual no flood	348	73
Actual flood	29	367

Both trained models were serialized with `joblib` and applied pixel-wise to the GeoTIFF exported from GEE (10 m scale), using `rasterio` to read the stack, masking invalid pixels, and writing the two binary flood maps as georeferenced single-band GeoTIFFs.

IV. RESULTS

A. Classification map comparison

Figure 2 compares the three flood-extent products: the SAR ratio-threshold mask (reference), the SVM classification and the Random Forest classification.

Visually, the two machine learning products are nearly identical to each other and reproduce the main flooded corridor along the Pantepec river detected by the SAR method, while extending the flooded class over additional low-lying agricultural areas at the margins of the SAR mask.

B. Total affected area

Table III summarizes the flooded surface estimated by each product (pixel size 10×10 m). The two optical classifiers estimate an affected area almost twice as large as the SAR threshold method.

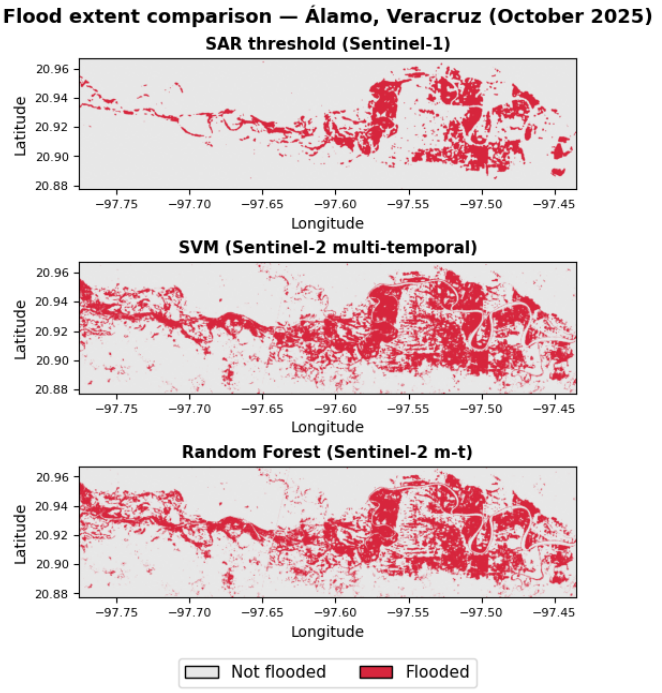


Fig. 2. Flood extent comparison over Álamo, Veracruz (October 2025): SAR threshold (Sentinel-1), SVM and Random Forest (Sentinel-2 multi-temporal).

TABLE III
FLOODED AREA PER PRODUCT

Product	Flooded pixels	Area (ha)	Coverage
SAR threshold (S-1)	495 745	4 957.5	13.06 %
SVM (S-2 multi-temp.)	950 683	9 506.8	25.04 %
Random Forest (S-2 m-t.)	953 424	9 534.2	25.11 %

V. CONCLUSIONS

Two supervised classifiers trained on multi-temporal Sentinel-2 features were able to reproduce, with validation accuracies above 82%, the flood mask produced by the UN-SPIDER SAR threshold methodology, while relying on a completely independent data source. The SVM proved slightly more stable than the Random Forest (cross-validated accuracy 0.869 ± 0.013 vs. 0.869 ± 0.016 probably due to the simplicity of the binary classification problem SVM with such a small feature set; which can be beneficial when using this approach in a computational systems with restricted resources or for larger areas of study.

Is also interesting to point at the fact that `B1`, `B12`, `MNDWI`, `NDWI` were chosen by both classifiers on the Sequential Feature Selection, which can indicate a further path of study with stronger methodologies to define the actual importance of each of the features selected. On the other hand, bands and indices `B2`, `B3`, `B4`, `B6`, `B7`, `B8`, `NDBI_diff`, `NDTI_diff` were not selected by any of the models, which can also show light into some improvements for future implementations excluding this information from the begining.

Several limitations must be also acknowledged. First, the ground truth itself is a proxy: the SAR threshold mask is a heuristic product with a manually tuned threshold, so label noise is embedded in the training data; an independent validation against high-resolution imagery or in-situ reports would be required to establish which product is closer to reality. Second, the optical method is inherently constrained by cloud cover: the post-event window had to be extended to 2025-10-15 to obtain usable acquisitions, so the optical maps may partially represent residual standing water or cloud coverage which could not be masked with traditional methods, rather than the flood peak, which could explain part of the larger detected extent. Third, permanent water bodies were labeled through the same change-detection logic, and masking with the JRC Global Surface Water dataset [3], as recommended by UN-SPIDER, is left as future refinement.

Overall, the dataset choice proved a useful use case for ranking and comparing supervised machine learning algorithms and the proposed pipeline can be considered a useful complement, rather than a replacement, to the SAR-based methodology: it provides flood maps when SAR is unavailable and an independent consistency check when it is. Future work includes validating against independent reference data, testing the transferability of the trained models to other flood events, adding auxiliary terrain features (DEM, slope, HAND, distance to river) and exploring feature selection and deep learning architectures.

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